

7. The glomerulus and Bowman’s capsule of the kidney nephron are the site of ultrafiltration.

Image 1 shows the main structures of this part of the nephron. **Image 2** shows an electron micrograph of part of the same structures.

Image 1

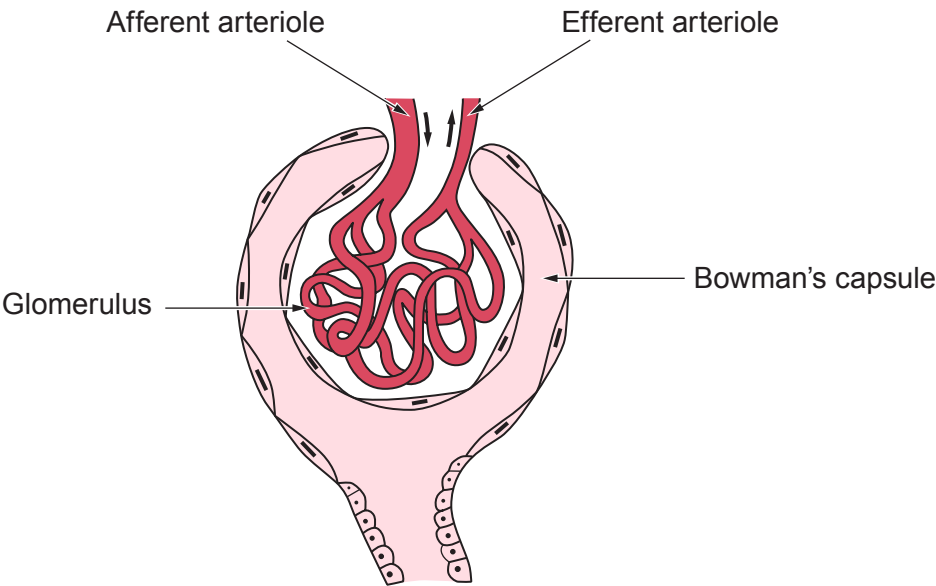
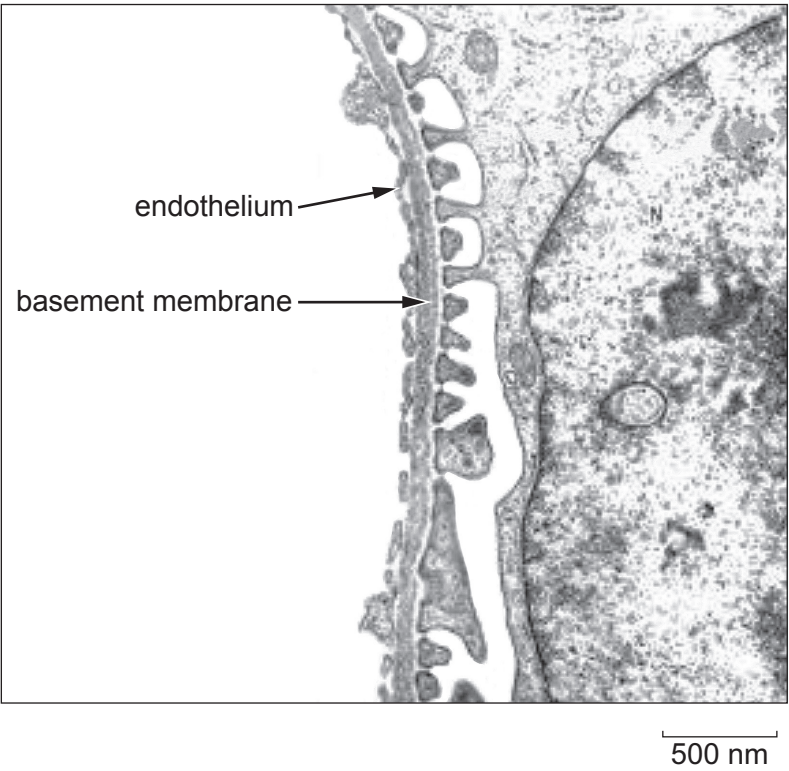


Image 2



Examiner
only

- (a) (i) On **Image 1**, label with a **P** the part of the nephron where the structures shown in **Image 2** would be found. [1]
- (ii) Using the scale bar on **Image 2** only, calculate the magnification of this electron micrograph. [2]

magnification = x

- (b) The effective pressure which forms the glomerular filtrate is due to several different pressures.

They are:

- the hydrostatic pressure in the glomerulus capillaries
- the osmotic pressure due to plasma proteins
- the intra-renal pressure (the pressure of the fluid already present in the Bowman's capsule and the tubules)

The pressures can be connected using the following formula:

$$\begin{array}{ccccccc} \text{pressure} & & & & & & \\ \text{forming} & = & \text{hydrostatic} & - & \text{osmotic} & - & \\ \text{glomerular} & & \text{pressure in} & & \text{pressure of} & & \text{intra-renal} \\ \text{filtrate} & & \text{glomerulus} & & \text{plasma} & & \text{pressure} \end{array}$$

In a healthy adult the osmotic pressure of the plasma is 4 kPa, the intra-renal pressure is 2.6 kPa and the hydrostatic pressure in the glomerulus is 8 kPa.

- (i) Calculate the pressure forming the glomerular filtrate. [1]

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- (ii) Use the information provided to suggest how a low protein diet could affect the rate at which glomerular filtrate is formed. [3]

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(iii) The pressure forming the glomerular filtrate can be regulated. The muscles in the walls of the afferent and efferent arterioles are under hormonal and nervous control.

Describe how changes in the afferent and efferent arterioles could increase the pressure forming the filtrate. [2]

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(c) In diseases such as diabetes mellitus and glomerulonephritis the membranes in the glomerulus and Bowman’s capsule are damaged and they become more permeable to proteins.

Suggest **two** reasons why the proteins are not usually reabsorbed back into the bloodstream. [2]

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(d) The urea concentration increases as the filtrate moves along the proximal convoluted tubule. However the mass of urea remains constant.

(i) Explain why the urea concentration increases. [1]

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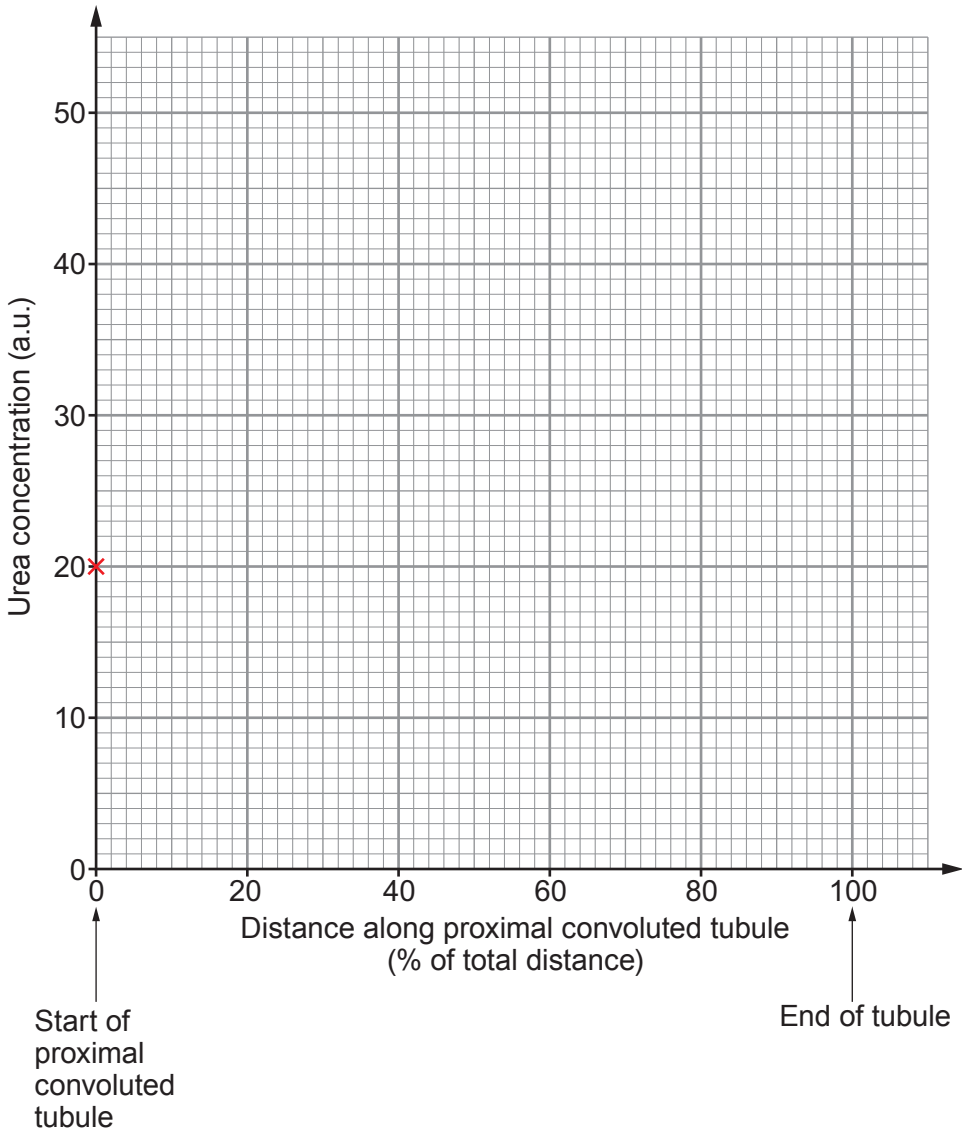


- (ii) If the concentration of urea along the proximal convoluted tubule is plotted on a graph, the line of best fit would follow the equation shown below:

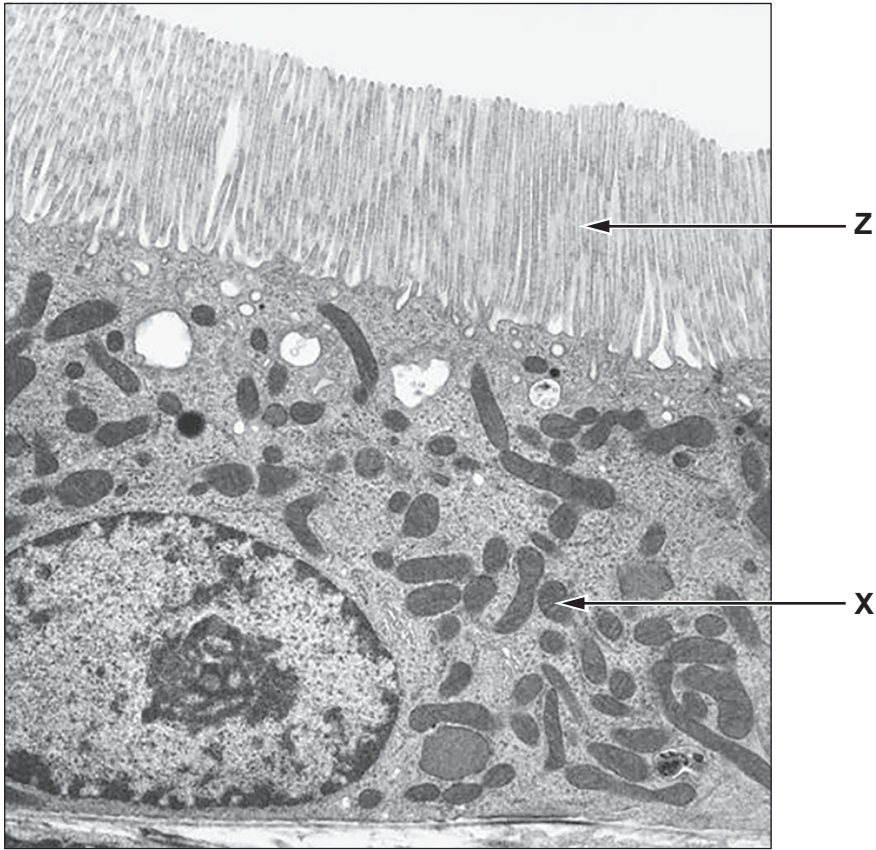
$$y = mx + c$$

The concentration of urea in the filtrate at the start of the proximal convoluted tubule is 20 arbitrary units (a.u.) and the value of m (the gradient) is 0.2 a.u./%.

Use the values provided to calculate the concentration of urea at the end of the tubule and draw a line on the axes below for the concentration of urea along the length of the proximal tubule. The concentration at the start of the proximal convoluted tubule has been plotted for you. [2]



(e) The electron micrograph below shows a section through the wall of the proximal convoluted tubule.



(i) Explain why there are many of the organelles labelled **X** in the cells shown. [2]

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(ii) Explain the presence of structures **Z** on the cell surface. [1]

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(f) Lungfish survive the drying out of their habitats by burrowing into the mud and decreasing their metabolic rate. Explain why, when in water, they release the products of deamination as ammonia but when they are in the dried-out mud, they convert the products to urea. [4]

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